Sanath Kumar Das

Chattogram, Bangladesh
Phone: +8801533648725
Email: sanathdas98@gmail.com
Portfolio — LinkedIn — Faculty Profile

Research Interests

- Brain–Computer Interfaces and Neural Decoding of Speech and Movement
- Deep Representation Learning for EEG and Physiological Signals
- Invasive and Non-invasive Neural Signal Processing
- Neuroprosthetics, Assistive Devices and Rehabilitation Robotics
- Closed-Loop Neurostimulation for Neurological Disorders

Education

Bangladesh University of Engineering and Technology (BUET)

Master's Degree in Electrical and Electronic Engineering *(Ongoing)*
CGPA: 3.63 / 4.00
Notable Courses: Biomedical Signal Processing, Medical Imaging, Machine Learning and Pattern Recognition, Deep Learning, Digital Image Processing

Bachelor's Degree in Electrical and Electronic Engineering 2018–2023
CGPA: 3.51 / 4.00
Major: Communication and Signal Processing

Publications

- **Sanath Kumar Das**, Arpan Sur, Md. Farhad Hossain. Collaborative Path Planning and Target Search in Multi-Robot Systems with PID-Controlled Uniform Area Partitioning. *2024 International Conference on Computer and Information Technology (ICCIT).* DOI: 10.1109/ICCIT64611.2024.11022590
- **Sanath Kumar Das**, Sudipta Saha, Fatin Fardaous, et al. Aquatic Robot for Trash Removal and Water Quality Monitoring Built from Recycled Materials. *2024 13th International Conference on Electrical and Computer Engineering (ICECE).* DOI: 10.1109/ICECE64886.2024.11024687

Research Experience

Master's Thesis (Ongoing) Supervisor: Dr. Shaikh Anowarul Fattah
Cross-Subject Generalization of Deep Learning Models for Decoding Inner Speech from EEG Signals

- Developing deep learning decoders for closed-vocabulary inner speech (imagined word) classification from multi-channel EEG, spanning person-specific models and cross-subject generalization with zero or few-trial calibration.
- Investigating domain adaptation, Euclidean/Riemannian alignment, and time–frequency, spatial, and connectivity-based features to reduce inter-subject variability.
- Implementing CNN, Transformer, hybrid, and graph neural network architectures to model spatial–temporal neural dynamics of EEG.
- Evaluating robustness, generalization, and neurophysiological interpretability (informative channels, frequency bands, time windows, connectivity) for real-world BCI applications.

Bachelor's Thesis Supervisor: Dr. Md. Farhad Hossain

A Collaborative Multi-Robot Coverage Path Planning and Efficient Target Search System

- Designed a cooperative multi-robot exploration system for unknown environments.
- Implemented coverage path planning and efficient target search algorithms.
- Performed simulation and timing analysis using CoppeliaSim and Python.

Teaching Experience

Lecturer, Department of Electrical and Electronic Engineering
East Delta University, Chattogram May 2024 – Present

- Prepare course materials including lecture notes, assignments, and exam questions.
- Deliver lectures and mentor undergraduate students for academic projects.

Technical Skills

Programming: Python, C, C++, MATLAB, Verilog, SystemVerilog
Machine Learning / Data Science: PyTorch, scikit-learn, NumPy, Pandas, Matplotlib
BCI Tools: MNE-Python, pyRiemann
Simulation / Tools: Simulink, Proteus, Quartus, CoppeliaSim

Academic Projects

Selected laboratory and course projects completed during undergraduate studies

- Design and hardware implementation of PPG-Based Pulse Monitoring and Counting System Using an IR sensor and amplifier
- Aquatic Multi-Purpose Robot for Water Surface Trash Cleaning and Water Quality Monitoring
- Simulation of 8-bit TDM-PCM Transmitter and Receiver System in Proteus
- Real-Time Human Detection and Social Distance Estimation Using YOLOv3
- IoT-Enabled Fire and Gas Leak Detection and Automatic Prevention System Using ESP32 Microcontroller

Awards and Achievements

- BUET RISE Research Grant 2022 for undergraduate thesis.
- Merit scholarships in the secondary and higher secondary examinations.

Leadership and Volunteer Activities

- General Secretary, BUET Robotics Society (2022–2023)
- Instructor at Basic Robotics Workshop organized by BUET Robotics Club

References

Dr. Shaikh Anowarul Fattah, Professor, Department of EEE, BUET
Phone: +8801937800854
Email: fattah@eee.buet.ac.bd

Dr. Md. Farhad Hossain, Professor, Department of EEE, BUET
Phone: +8801935591028
Email: mfarhadhossain@eee.buet.ac.bd